Andres Mosquera Hernandez

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Education

- **Universidad de los Andes** Bogota, Colombia
  M.Sc. (research) in Systems and Computing Engineering *Aug 2025 – Dec 2027 (expected)*
  - **Scholarship**: Full scholarship (Beca Avanza) as a graduate research assistant.
  - **Thesis**: Process mining and mechanistic interpretability for explaining prescriptive process monitoring.
- **BlueDot Impact** Remote
  Technical AI Safety Course and Technical AI Safety Project *2026*
  - **Project**: Extended my Apart hackathon project on chain-of-thought bias attribution into a workshop paper, mentored by Jess Bergs (UK AI Security Institute).
  - **Course**: Alignment, mechanistic interpretability, evaluations, AI control, and scalable oversight. Also completed Future of AI.
- **Uniempresarial – Chamber of Commerce of Bogota** Bogota, Colombia
  B.Sc. in Software Engineering, summa cum laude; GPA: 4.6/5 *Aug 2021 – March 2025*
  - **Awards**: Ranked 1st in graduation class. 7x Best Course Programming Project. Highest Saber Pro score (national exam) in the program.
  - **Research**: AI teaching assistant. Developed PhingPT, an LLM for the financial sector, at the National University of Colombia.

Research and Publications

- **Thought Anchors for Social Bias: Which Reasoning Steps Matter in Extended Thinking LLMs on Latin American Scenarios**: **A. Mosquera-Hernandez**. Winner, Global South AI Safety Hackathon (Apart Research, 2026; competing solo, 217 teams). Submitted to GeBNLP @ ACL 2026.
  - Located where stereotypes enter LLM reasoning across 1,708 chunk-level attributions on 3 extended-thinking models (Claude Sonnet 4.6, Gemini 2.5 Flash, Qwen3.7-Plus), by adapting Thought Anchors resampling to the SESGO benchmark (4 bias categories, 2 languages).
  - Found that high-impact steps are general inference and context recall, not overt stereotype statements. Shipped pipeline, paper, and interactive demo in about 24 hours.
- **Rare Once It Costs Anything: Costly Cooperation Between LLM Agents**: D. J. Daza Jaimes, **A. Mosquera-Hernandez**, V. M. Gelves Cabrera. Apart Research AI Incident Response Sprint (Sep 2026). Top 25% of 230+ projects.
  - Tested whether tool-using LLM agents pay from their own task budget to help a stranger (127 runs, 6 agents each). Found that once helping costs anything, few agents pay (23.9% vs. about 47% when free).
  - Built a hash-chained run ledger and a preregistered contrast; reviewers noted every number reproduces from the repository.
- **How Multiple Explanations Behind a Decision Compose in Prescriptive Process Monitoring**: **A. Mosquera-Hernandez**, O. González-Rojas. Under review at ICPM (International Conference on Process Mining).
  - Designed RETIME, which composes the explanations of the risk, effect, and timing models behind each recommendation. Showed that the effect model carries 56–81% of the decision's reasons, validated with deletion tests and path patching on BPIC2012 and BPIC2017.
- **Automatic Legal Judgment Summarization Using Large Language Models: A Case Study for the JUST-NLP 2025 Shared Task**: **A. Mosquera-Hernandez** et al. Under review (journal).
  - Placed 3rd on the official JUST-NLP 2025 leaderboard by summarizing 400 unseen Indian court judgments with GPT-4.1 prompting, hybrid extractive–abstractive pipelines, and multi-agent state graphs (ROUGE-2, ROUGE-L, BLEU).

Experience

- **Universidad de los Andes** — Bogota, Colombia
  Graduate Research Assistant — *Aug 2025 - Present*
  - **Research**: Explainable prescriptive process monitoring with ML, RL, causal inference, and LLMs (Python, PyTorch).
  - **Service**: Poster evaluator for IEEE-affiliated academic events.
- **AI Safety Colombia** — Bogota, Colombia
  Facilitator, Technical AI Safety Seminar (BlueDot Impact curriculum) — *Oct 2026 - Nov 2026*
  - **Teaching**: Facilitating a 6-week seminar in Spanish for a cohort of 8 on safer training, evaluations, interpretability, and harm mitigation.
- **Coplanin** — Bogota, Colombia (Remote)
  Machine Learning Engineer / Full Stack / Android Engineer — *April 2023 - June 2025*
  - **Vision**: Cut manual rework by 60% by training a Pix2Pix GAN for cadastral image segmentation (0.87 IoU, 92% accuracy).
  - **LLMs**: Split 57,000 Spanish names (81% accuracy) and inferred gender (95%) by fine-tuning Gemma-2, Llama 3, Mistral, Phi-4, and Qwen2.5 (Unsloth).
  - **Data**: Classified 35,000 addresses (93% accuracy). Built an Airflow ETL for the Colombian Ministry of Environment, 3x faster than manual.
  - **Apps**: Shipped property-management and water-network apps (Kotlin, React, Next.js), one with 700 users. Led two engineers.
- **Kohl's** — Remote
  Machine Learning Ops Intern — *Aug 2024 - Oct 2024*
  - **MLOps**: Reduced model deployment time by 73% on 10M-record datasets by automating pipelines (Kubernetes, Vertex AI, Airflow).
- **Universidad del Rosario** — Bogota, Colombia
  Researcher — *Jun 2025 - Jul 2025*
  - **AR**: Saved USD 2,000 in 3D printing by building Unity AR labs where 100+ students inspect bone models before printing.
- **Alpha Intelligence AI** and **Corparques** — Bogota, Colombia
  Software Engineer Intern — *May 2022 - Jan 2023*
  - **Software**: Analyzed a 10 TB judicial-branch database (Python, SQL, Azure). Cut operational task time by 87% with automation.

Awards and Talks

- **Winner – Global South AI Safety Hackathon (Apart Research, 2026)**: Competing solo among 217 teams. USD 1,000 prize.
- **Speaker – International Symposium of Statistics (Pasto, Colombia, 2025)**: Financial sentiment analysis with LLMs (RAG, fine-tuning).
- **1st Place – AI Learning Tools Challenge, Scotiabank Colpatria (2024)**: Among 32 universities.

Skills

- **ML and research**: Python, PyTorch, Hugging Face, scikit-learn, NumPy, Pandas, LangChain, Unsloth.
- **Engineering**: Docker, Kubernetes, Google Cloud, AWS, Airflow, Git, Linux, SQL, TypeScript, Java, C++.
- **AI tools**: Claude Code, Codex, Cursor.
- **Languages**: Spanish (native), English (professional working proficiency, B2).